

Differences in the constitutive terpene profile of lodgepole pine across a geographical range in British Columbia, and correlation with historical attack by mountain pine beetle

Erin L. Clark, Allan L. Carroll, Dezene P.W. Huber

Abstract—The mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae), is a destructive insect pest in western Nearctic conifer forests. Currently, British Columbia, Canada, is experiencing the largest recorded outbreak of this insect, including areas that historically have had low climatic suitability for it. We analyzed 26 constitutive resin terpenes in phloem samples from British Columbia lodgepole pine (*Pinus contorta*) populations to test for differential resistance to mountain pine beetle attack, based upon the likelihood of previous exposure to mountain pine beetle. We assessed sampled trees for number of mountain pine beetle attacks, number of pupal chambers, and tree survival the following spring. Significant differences were found when levels of certain terpenes in lodgepole pine populations that had likely experienced substantial mountain pine beetle infestations in the past were compared with those in populations that likely had not experienced large outbreaks of mountain pine beetle. Although we expected southern pine populations to contain more total terpenes than northern populations, owing to higher historical exposure to the beetle, the converse was found. Northern populations generally had higher levels of constitutive terpenes and beetle attack than southern populations. Because several terpenes are kairomones to the mountain pine beetle and also serve as precursors for the synthesis of pheromones, the lower levels of terpenes expressed by lodgepole pines from the historical range of the mountain pine beetle may render them less chemically perceptible to foraging beetles.

Résumé—Le dendroctone du pin ponderosa, *Dendroctonus ponderosae* Hopkins (Coleoptera : Curculionidae), est un insecte ravageur des forêts de conifères de l'Ouest néarctique. À présent, la Colombie-Britannique connaît l'épidémie la plus importante signalée de cet insecte, même dans des régions dont le climat dans le passé a été peu propice à ce coléoptère. Nous avons analysé 26 terpènes constitutifs dans des échantillons de phloème de populations de pins vrillés (*Pinus contorta*) de la Colombie-Britannique pour tester leurs différences de résistance aux attaques du dendroctone du pin ponderosa basées sur la probabilité d'expositions antérieures au dendroctone du pin ponderosa. Nous avons déterminé chez les arbres échantillonnés le nombre d'attaques par le dendroctone du pin ponderosa, le nombre de chambres nymphales et la survie des arbres au printemps suivant. Il existe des différences significatives de concentrations de certains terpènes entre les populations qui ont vraisemblablement connu d'importantes infestations de dendroctones du pin ponderosa dans le passé et les populations qui n'ont pas subi de telles grandes infestations du dendroctone du pin ponderosa. Bien que nous nous attendions à ce que les populations du sud contiennent plus de terpènes totaux que les populations du nord à cause de leur plus forte exposition dans le passé aux coléoptères, c'est l'inverse qui prévaut. Les populations nordiques ont généralement de plus fortes concentrations de terpènes constitutifs et de plus forts taux d'attaques des coléoptères que les populations du sud. Parce que plusieurs des terpènes sont des kairomones du dendroctone du pin ponderosa et servent de précurseurs dans la synthèse des phéromones, les concentrations plus faibles de terpènes affichées par les pins vrillés dans l'aire de répartition historique du dendroctone du

Received 22 December 2009. Accepted 18 June 2010.

Erin L. Clark,¹ Dezene P.W. Huber, Ecosystem Science and Management Program, University of Northern British Columbia, 3333 University Way, Prince George, British Columbia, Canada V2N 4Z9
Allan L. Carroll, Forest Sciences Department, University of British Columbia, 2424 Main Mall, Vancouver, British Columbia, Canada V6T 1Z4

¹Corresponding author (e-mail: eclark1@unbc.ca).
doi: 10.4039/n10-022

pin ponderosa peut les rendre chimiquement moins perceptibles aux coléoptères en recherche de nourriture.

[Traduit par la Rédaction]

Introduction

One of the primary means of defense utilized by coniferous trees against insect attack is resin, which is composed primarily of terpenes. Resin provides both a physical and a chemical defense and can be both a constitutive and an inducible defense. Constitutive defenses are maintained by a tree at all times for defense from initial attacks, whereas induced defenses are initiated when a tree faces a threat (Franceschi *et al.* 2005). In the study we focused on the constitutive defense provided by terpenes. Bark beetles (Coleoptera: Curculionidae: Scolytinae) may exhibit positive or negative chemotactic responses to terpenes when selecting a host (Gershenzon and Croteau 1991; Keeling and Bohlmann 2006; Pureswaran *et al.* 2006). Some terpenes are toxic to invaders (Smith 1963, 1965). Despite this, bark beetles, including the mountain pine beetle, *Dendroctonus ponderosae* Hopkins, are able to use some resin terpenes in the host phloem as precursors to aggregation-pheromone components (Hughes 1973; Raffa and Berryman 1983b; Seybold *et al.* 2006).

The mountain pine beetle is a native bark beetle normally present at endemic levels throughout western North America from Mexico to northern British Columbia. It is a primary insect herbivore in lodgepole pine (*Pinus contorta* Douglas *ex* Loudon var. *latifolia* Engelm. *ex* S. Watson (Pinaceae)) ecosystems (Amman and Cole 1983) and has been described as the most destructive of the tree-killing species of *Dendroctonus* Erichson (Craighead *et al.* 1931). Newly emerged adults search for host trees in midsummer; adult females bore through the outer bark and construct galleries in the phloem. Because reproductive success is dependent on their ability to kill their hosts (Raffa and Berryman 1983b; Logan and Powell 2001), the beetles produce aggregation pheromones to attract conspecifics; the resultant mass attack by beetles can overwhelm host-tree defenses (Raffa and Berryman 1983b; Borden 1985). The rapid

recruitment of both sexes of conspecifics serves to increase the levels of pheromone and host-derived kairomones emanating from a tree, and attack rapidly escalates until the tree succumbs. In addition to their tunneling activity in the phloem, the beetles vector pathogenic fungi into the tree (Reid *et al.* 1967; Lee *et al.* 2006).

Endemic mountain pine beetle populations persisting in physiologically weakened trees occasionally reach outbreak proportions and begin to attack and kill apparently healthy trees at the landscape level (Rudinsky 1962; Safranyik and Carroll 2006). Four or five significant mountain pine beetle outbreaks occurred in British Columbia, Canada, in the 20th century (Taylor and Carroll 2004). From 1999 to 2006, approximately 13 million ha were affected in British Columbia (Westfall and Ebata 2008), the largest outbreak on record (Carroll *et al.* 2004). In addition, that infestation extended farther north in the province and to higher elevations than previously recorded (Carroll *et al.* 2004; Raffa *et al.* 2008).

Conifer terpene composition is a heritable trait (Hanover 1966; Squillace 1971) controlled by numerous terpene synthase genes (Keeling and Bohlmann 2006). Within a population of conifers (including lodgepole pine), there is considerable variation in resin monoterpene composition (Hanover and Furniss 1966; Franklin and Snyder 1971; Squillace 1971; Rockwood 1973; Forrest 1980; Smith 1983).

Both constitutive and induced terpene-based defenses may have evolved partly in response to selective pressure by bark beetles (Mitton and Sturgeon 1982; Raffa and Berryman 1987). Lodgepole pine may also show within- and between-population variation in chemical defenses that could translate into variable levels of susceptibility to beetle attack and be manifested by higher levels of attack in new areas during times of range expansion by the mountain pine beetle. Raffa and Berryman (1982) found no differences in percent composition of the monoterpenes present in lodgepole pine resistant and susceptible to mountain

pine beetle attack. They also found compositional changes but no qualitative difference in the induced phloem chemical composition between resistant and susceptible trees. They concluded that the quantitative resin response was greater in resistant trees after attack, but that monoterpene composition did not differ between resistant and susceptible lodgepole pines. However, that study took place in one location, and consequently did not address variation potentially present at a larger geographical scale. Plasticity of the multigene-based terpene defense in conifers is an important adaptation for these long-lived, immobile species in their defense against shorter lived, highly mobile herbivores and pathogens (Huber *et al.* 2004). This implies that trees in areas with prior exposure to bark beetle outbreaks may be better adapted to defending themselves against beetles. We predicted that pines in regions that have historically been exposed to high mountain pine beetle pressure will have evolved more effective terpene defenses and responses against beetles than pines in regions with lower historical exposure to beetle pressure.

The results of other studies have supported this prediction. Although Hanover and Furniss (1966) found no qualitative difference in monoterpene composition between Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco (Pinaceae)) trees attacked and not attacked by Douglas-fir beetle (*Dendroctonus pseudotsugae* Hopkins), they did find higher levels of Δ -3-carene in unattacked trees than in those that resisted attack, which implies that particular terpenes played a role in resistance in some trees. Sturgeon (1979) found that ponderosa pines (*Pinus ponderosa* C. Lawson) from areas that had historically been exposed to outbreaks of western pine beetle (*Dendroctonus brevicomis* LeConte) had higher concentrations of limonene. Smith (1975) found that ponderosa pines with a higher percentage of limonene and higher resin flow were more resistant to attack and colonization by western pine beetle than trees with lower limonene levels or resin flow.

We used a climatic-suitability model for mountain pine beetle (Carroll *et al.* 2004) to infer probable historical beetle pressure, based upon the suitability of historical climatic

factors for sustaining beetle populations. The model is based upon climatic factors that are known to be important to mountain pine beetle success, such as day-degree accumulation, minimum winter temperatures, and amount of spring precipitation. Particular geographical regions are placed in climatic-suitability classes (CSCs), using a low-to-high ranking system. This model allows selection of sampling sites in areas hypothesized to have historically been exposed to different levels of mountain pine beetle pressure.

To assess whether the constitutive resin defenses of lodgepole pine that could relate to susceptibility to attack by mountain pine beetle in regions that likely had historically high exposure to the beetle differed from those in regions that likely had not, we sampled pines across a geographical range and among different CSCs in British Columbia. We predicted that there would be differences in constitutive terpene composition and level of tree attacks between populations that correlate with historical climatic suitability for mountain pine beetle attack and with subsequent beetle reproductive success.

Methods

Sample collection and processing

Four locations were selected on the basis of accessibility and geographical spread along an approximately 520 km north–south transect across the mountain pine beetle infestation zone in the interior of British Columbia (Table 1). Two sites were selected at each of the most northerly (Quesnel) and southerly (Princeton) locations: one site in a historically climatically favorable area (high CSC) for mountain pine beetle and one in a less favorable (low CSC) area, based on climate normals for the period 1941–1970 (Carroll *et al.* 2004). Sites in both high and low CSCs with unattacked trees were found near Princeton and sites with severe and moderate CSCs and uninfested trees were found near Quesnel. Near 100-Mile House and Kamloops, only one site each was selected in the low CSC (because of the high infestation level in those areas, suitably sized, unattacked trees in the high CSC could not be found). Only uninfested lodgepole pine trees (green foliage present, no pitch

Table 1. Sampling locations, sites (representing climatic-suitability classes (CSC)), numbers of trees sampled, and CSCs of the sampled plots of lodgepole pine (*Pinus contorta*) trees in British Columbia.

Location	Site	No. of plots	No. of trees sampled	Coordinates	CSC*
Quesnel	Higher	7	66	53°00.555'N, 122°11.941'W	Severe
	Lower	10	56	53°08.827'N, 121°51.741'W	Moderate
100 Mile House	Lower	6	79	51°47.587'N, 120°45.372'W	Moderate
Kamloops	Lower	3	55	50°31.900'N, 120°32.102'W	Very low
Princeton	Higher	8	59	49°48.339'N, 120°31.743'W	High
	Lower	4	51	49°49.597'N, 120°27.534'W	Low

*Carroll *et al.* (2004).

tubes or frass on stems) were baited and (or) sampled. At each site, three to ten 7 m fixed-radius plots were established (Table 1). In each plot, a central tree with DBH ≥ 15 cm was selected for baiting with mountain pine beetle lure (Contech, Inc., Delta, British Columbia). In two cases it was necessary to hang the central tree bait on a lodgepole pine with DBH < 15 cm. All uninfested, mature (DBH > 15 cm) experimental pines within each plot were measured and labeled with tree-marking paint (Aervoe Industries Inc., Gardnerville, NV 89410, United States of America) and their DBH and distance from the central bait were measured. A minimum distance of 25 m was maintained between the centers of adjacent plots. The number of plots varied among sites, but enough plots were established at each site to obtain a minimum of 50 experimental trees. Baiting and initial sampling at all locations were completed between 6 and 12 July 2006, before the major beetle flight that year.

Immediately after baiting, a phloem and bark sample was taken from each tree in each plot 1.3 m from ground level using a 10 mm diameter punch (C.S. Osborne & Company, Harrison, New Jersey). Each phloem and bark disk was stored in an individual 5.7 cm $\times$ 8.9 cm envelope and immediately placed on dry ice until transfer to a -80°C freezer (Thermo Fisher Scientific, Waltham, Massachusetts) in the laboratory. All disks were

stored at -80°C until shipment to the British Columbia Ministry of Forests and Range Forest Research Laboratory, Victoria, British Columbia, for processing and analysis.

During 20–24 August, at the end of the major beetle flight, entrance holes within a 20 cm $\times$ 40 cm frame placed at approximately 1.6 m height on the east and west sides of each attacked tree in each plot were counted to estimate mountain pine beetle attack density (number of entrance holes/m²). In May 2007, the experimental trees in the plots were revisited to determine pupal-chamber density as a measure of beetle reproductive success on the trees. A 10 cm $\times$ 20 cm rectangle was outlined on the east and west sides of each attacked tree at DBH. The bark was peeled off and the pupal chambers in each rectangle were counted. The totals were then used to calculate pupal-chamber density (number/m²). Between these sampling periods, woodpeckers or other animals had removed the bark of several experimental trees. Thus, it could not be determined in all cases whether lower pupal-chamber numbers were due to predation or to larval death following bark removal. In addition, several replicates were no longer available for assessment in the spring of 2007 because of logging activity. Therefore, data were collected from only a subset of the original sampling group, somewhat reducing the data set, particularly at the northernmost site.

Phloem samples were processed using gas chromatographic – flame ionization detection analyses to identify compounds by matching their retention time with those of synthetic standards at the British Columbia Ministry of Forests and Range laboratory. Frozen (-80°C) phloem samples were ground in liquid nitrogen, then extracted in 4 mL of hexane (with 250 ppm pentadecane as an internal standard) for 48 h. Samples were then inverted and allowed to settle for 24 h, after which 0.5 mL of solution was transferred to a 2 mL autosampler vial for gas chromatographic analysis using either a PerkinElmer Clarus500 or PerkinElmer AutoSystem (PerkinElmer, Waltham, Massachusetts) with built-in autosampler, fitted with an INNOWax column (J&W, 25 m $\times$ 0.2 mm i.d., 0.4 μm film). The injection was split (35 mL/min, approximately 39:1; injector temperature 200°C). Helium was utilized as the carrier gas (flow rate 21 psi, 0.90 mL/min at 60°C). The oven temperature was held at 60°C for 1 min, increased at a rate of $3.0^{\circ}\text{C}/\text{min}$ to 85°C , then increased at a rate of $8.0^{\circ}\text{C}/\text{min}$ to 170°C . Finally, the temperature was increased by $20.0^{\circ}\text{C}/\text{min}$ to 250°C and held for 7.0 min. The remaining contents of the vials containing the extracted phloem and bark were evaporated in a fume hood and then oven-dried at 70°C overnight to remove residual moisture. The dry mass of each sample was obtained to determine a moisture correction, which was applied to the results.

All data were analyzed using R version 2.6.2 (R Development Core Team 2008). Negligible values of < 5 ppm were considered to be zero for analysis. For all tests in which differences between locations were examined, only samples from the lower CSC sites at all four locations were considered. Tests examining the differences between CSCs were completed using data derived from the higher CSC and lower CSC sites in the northern- and southern-most sampling locations.

Variation in terpene levels by location and CSC

Data were checked for assumptions of homogeneity of variance and normality using Levene's (Mickey *et al.* 2004) and Shapiro–Wilk tests, respectively. Phloem resin terpene data that met both assumptions were analyzed

using ANOVA ($\alpha = 0.05$) followed by a multiple comparison of means with Tukey's HSD test ($\alpha = 0.05$). Data that could not be transformed to meet the assumptions of homoscedasticity and normality were analyzed using a Kruskal–Wallis rank-sum test followed by a pairwise Wilcoxon's rank-sum test ($\alpha = 0.05$).

Effects of terpenes on attack density and beetle reproductive success

Differences in attack density between locations and between CSCs were analyzed using ANOVA ($\alpha = 0.05$) followed by a multiple comparison of means with Tukey's HSD test ($\alpha = 0.05$) to determine differences. Square-root transformations were necessary to meet assumptions for these tests, based upon a visual examination of residual plots for the comparisons between locations and between CSCs.

A linear mixed-effects model was used to examine the effects of specific terpenes and geographical variation on attack density and pupal-chamber density. The model designed to examine effects on attack density was created using a combination of backward elimination and forward addition, with the seven terpenes that made up the majority of the resin terpenes, eliminating the highest P values until all variables remaining in the model were significant ($\alpha = 0.05$). The final model contained a number of fixed effects: DBH, location, and limonene concentration. The random effect was plot. No transformation was required to meet the assumptions of normality and homoscedasticity, based upon a visual inspection of the residual plots. The model examining the variation in attack density by CSC at the northern- and southern-most sampling sites showed that location, CSC, and DBH were fixed effects and plot was the random effect in the final model.

The model for examining effects on pupal-chamber density was created using a combination of backward elimination and forward addition, eliminating the highest P values until all variables remaining in the model were significant ($\alpha = 0.05$). No transformation was necessary to meet the assumption of normality, based upon a visual examination of the data. In addition, there was a nonlinear relationship between attack density and number of pupal chambers. This was expected because

intraspecific competition for resources should have reduced the success rate of larvae before they had a chance to pupate, regardless of host suitability. This outcome was accounted for in the model. The model contained fixed effects of location, DBH, and attack density, with random effect of plot.

Effects of terpenes on tree attack status

Trees were also reassessed in spring 2007 to determine survival. Each tree was categorized into a binomial survival class (green = alive, red/fading = dead) and a combination of backward elimination and stepwise addition was used to create a binomial generalized linear model with random effect of plot, in which all terms were significant ($\alpha = 0.05$).

Results

Variation in terpene levels between locations and CSCs

There were significant differences in the amounts of almost all terpenes between locations (Table 2). The northernmost site, Quesnel, often contained significantly higher amounts of the terpenes identified than the other three sites. Levels of total terpenes were also significantly ($F_{3,236} = 109$, $P < 0.001$) greater at that site. Most individual terpenes comprised $\leq 1\%$ of total terpenes present in trees, but seven predominated and comprised, on average, $> 1\%$ of total terpenes: Δ -3-carene, limonene, myrcene, β -phellandrene, α -pinene, β -pinene, and terpinolene. The seven predominant terpenes showed similar trends across the sampled geographical area, their levels at Quesnel being significantly higher (β -phellandrene: $F_{3,236} = 91.9$, $P < 0.001$; β -pinene: $\chi^2_3 = 75.7$, $P < 0.001$; Δ -3-carene: $\chi^2_3 = 77.5$, $P < 0.001$; limonene: $\chi^2_3 = 66.5$, $P < 0.001$; α -pinene: $\chi^2_3 = 105.5$, $P < 0.001$; myrcene: $\chi^2_3 = 108.4$, $P < 0.001$; terpinolene: $\chi^2_3 = 64.8$, $P < 0.001$) than elsewhere (Table 2).

A comparison between the higher and lower CSCs at Quesnel and Princeton showed significant differences within each location in the levels of terpene compounds (β -phellandrene: $\chi^2_3 = 81.2$, $P < 0.001$; β -pinene: $F_{3,227} = 17.59$, $P < 0.001$; Δ -3-carene: $\chi^2_3 = 50.5$, $P < 0.001$; limonene: $\chi^2_3 = 53.0$, $P < 0.001$; α -pinene: $\chi^2_3 = 68.2$, $P < 0.001$; myrcene: $\chi^2_3 = 89.8$,

$P < 0.001$; terpinolene: $\chi^2_3 = 33.5$, $P < 0.001$; α -phellandrene: $\chi^2_3 = 75.8$, $P < 0.001$; sabinene: $\chi^2_3 = 71.3$, $P < 0.001$; bornyl acetate: $\chi^2_3 = 21.4$, $P < 0.001$; camphene: $\chi^2_3 = 70.2$, $P < 0.001$; linalool: $\chi^2_3 = 91.3$, $P < 0.001$; terpineol: $\chi^2_3 = 89.0$, $P < 0.001$; ocimene: $\chi^2_3 = 21.2$, $P < 0.001$; α -humulene: $\chi^2_3 = 46.9$, $P < 0.001$; α -copaene: $\chi^2_3 = 49.7$, $P < 0.001$; borneol: $\chi^2_3 = 30.9$, $P < 0.001$; α -caryophyllene: $\chi^2_3 = 53.2$, $P < 0.001$; pulegone: $\chi^2_3 = 13.3$, $P = 0.004$; *p*-cymene: $\chi^2_3 = 14.27$, $P = 0.003$; γ -terpinene: $\chi^2_3 = 33.4$, $P < 0.001$; camphor: $\chi^2_3 = 66.7$, $P < 0.001$; α -terpinene: $\chi^2_3 = 9.2$, $P = 0.027$; α -cubebene: $\chi^2_3 = 4.8$, $P = 0.184$; total terpenes: $\chi^2_3 = 90.8$, $P < 0.001$). There were also significant differences between locations, although these differences were not as consistent (Table 3). The higher CSC sites in Quesnel and Princeton were often not significantly different from each other in terms of particular terpenes. The lower CSC site at the northernmost site consistently had the highest levels of the seven predominant terpenes, whereas the lower CSC site at the southernmost site consistently had the lowest concentrations of the same compounds (Table 3).

Effects of terpenes on attack density and beetle reproductive success

There were significant differences ($F_{3,236} = 19.51$, $P < 0.001$) in attack density between the four locations (Fig. 1). Mean attack density was statistically higher at the northernmost sampling sites than at the southernmost sampling sites. There was no significant difference in mean attack density per tree between Quesnel and 100-Mile House or between Kamloops and Princeton.

Attack density was influenced by several of the measured parameters. Location, DBH, and limonene concentration were all significant factors (location: $F_{3,225} = 19.29$, $P < 0.001$; DBH: $F_{1,225} = 18.15$, $P < 0.001$; limonene concentration: $F_{1,225} = 6.75$, $P = 0.01$) in modeling attack density (Table 4; see Appendix A (Table A1) for the correlation between terpenes). Increases in DBH and limonene concentration were related to an increase in attack density (Table 4). Location, DBH, and attack density played a significant role in larval success as measured by pupal-chamber

Table 2. Amounts of terpenes present (ppm, mean $\pm$ 1 SE) for constitutive defense in lodgepole pine (*Pinus contorta*) trees by location in British Columbia (Quesnel site (lower CSC), 100 Mile House site, Kamloops site, and Princeton site (lower CSC)).

Terpene	Location				$F_{3,236}$	P
	Quesnel ($n = 56$)	100 Mile House ($n = 79$)	Kamloops ($n = 55$)	Princeton ($n = 50$)		
β -Phellandrene	17 869.5 $\pm$ 972.5a	6266.4 $\pm$ 468.9b	4320.8 $\pm$ 397.0c	5538.9 $\pm$ 457.4bc	$F_{3,236} = 91.9$	$P < 0.001$
β -Pinene	2 598.7 $\pm$ 307.1a	584.1 $\pm$ 75.9b	410.8 $\pm$ 53.4b	623.1 $\pm$ 78.0b	$\chi^2_3 = 75.7$	$P < 0.001$
Δ -3-Carene	1 946.6 $\pm$ 285.7a	580.3 $\pm$ 122.6b	223.6 $\pm$ 48.6c	517.6 $\pm$ 146.4b	$\chi^2_3 = 77.5$	$P < 0.001$
Limonene	1 891.9 $\pm$ 350.9a	502.3 $\pm$ 90.0b	565.5 $\pm$ 75.7c	335.2 $\pm$ 53.1b	$\chi^2_3 = 66.5$	$P < 0.001$
α -Pinene	1 366.7 $\pm$ 147.3a	395.8 $\pm$ 31.6b	276.5 $\pm$ 20.6c	363.2 $\pm$ 28.5bc	$\chi^2_3 = 105.5$	$P < 0.001$
Myrcene	827.0 $\pm$ 62.2a	215.3 $\pm$ 25.3b	177.5 $\pm$ 17.8b	211.4 $\pm$ 16.9b	$\chi^2_3 = 108.4$	$P < 0.001$
Terpinolene	406.6 $\pm$ 40.1a	131.2 $\pm$ 17.5b	130.3 $\pm$ 20.1b	138.3 $\pm$ 19.7b	$\chi^2_3 = 64.8$	$P < 0.001$
α -Phellandrene	306.4 $\pm$ 18.0a	91.7 $\pm$ 8.6b	73.1 $\pm$ 7.4b	92.6 $\pm$ 8.7b	$\chi^2_3 = 106.3$	$P < 0.001$
Sabinene	260.4 $\pm$ 19.2a	88.8 $\pm$ 7.9b	55.1 $\pm$ 6.7c	78.9 $\pm$ 8.0b	$\chi^2_3 = 104.1$	$P < 0.001$
Bornyl acetate	240.0 $\pm$ 81.5	70.6 $\pm$ 14.6	60.6 $\pm$ 4.5	59.5 $\pm$ 5.2	$\chi^2_3 = 0.3$	$P = 0.956$
Camphene	179.9 $\pm$ 29.0a	46.9 $\pm$ 5.1b	31.9 $\pm$ 3.9b	42.3 $\pm$ 3.6b	$\chi^2_3 = 105.8$	$P < 0.001$
Linalool	130.8 $\pm$ 13.6a	16.9 $\pm$ 2.3b	10.4 $\pm$ 2.7c	16.4 $\pm$ 3.3bc	$\chi^2_3 = 123.8$	$P < 0.001$
Terpineol	124.7 $\pm$ 10.2a	25.9 $\pm$ 4.8b	16.3 $\pm$ 3.0b	23.4 $\pm$ 5.2b	$\chi^2_3 = 100.6$	$P < 0.001$
Ocimene	123.8 $\pm$ 33.1a	15.7 $\pm$ 4.8b	22.4 $\pm$ 5.8bc	33.8 $\pm$ 8.6c	$\chi^2_3 = 46.2$	$P < 0.001$
α -Humulene	93.9 $\pm$ 12.1a	31.3 $\pm$ 6.5b	16.3 $\pm$ 4.0b	16.7 $\pm$ 4.3b	$\chi^2_3 = 57.7$	$P < 0.001$
α -Copaene	51.7 $\pm$ 6.8a	4.5 $\pm$ 1.3bc	2.5 $\pm$ 1.6b	8.0 $\pm$ 2.3c	$\chi^2_3 = 91.5$	$P < 0.001$
Borneol	51.6 $\pm$ 18.9a	7.6 $\pm$ 1.9b	3.1 $\pm$ 0.9c	5.6 $\pm$ 1.1bc	$\chi^2_3 = 28.8$	$P < 0.001$
α -Caryophyllene	46.0 $\pm$ 6.1a	7.9 $\pm$ 2.2b	3.7 $\pm$ 1.2b	8.8 $\pm$ 2.7b	$\chi^2_3 = 90.2$	$P < 0.001$
<i>p</i> -Cymene	31.5 $\pm$ 4.7a	26.1 $\pm$ 5.7a	11.1 $\pm$ 2.7b	10.6 $\pm$ 2.1b	$\chi^2_3 = 20.8$	$P < 0.001$
Pulegone	30.3 $\pm$ 5.4a	9.4 $\pm$ 2.1b	6.1 $\pm$ 1.7b	8.0 $\pm$ 1.8b	$\chi^2_3 = 27.6$	$P < 0.001$
γ -Terpinene	27.1 $\pm$ 3.7a	3.3 $\pm$ 1.4b	2.1 $\pm$ 0.9b	3.9 $\pm$ 1.6b	$\chi^2_3 = 69.2$	$P < 0.001$
Camphor	15.0 $\pm$ 3.6b	31.1 $\pm$ 2.7a	9.9 $\pm$ 1.9b	2.1 $\pm$ 0.7c	$\chi^2_3 = 70.8$	$P < 0.001$
α -Terpinene	14.5 $\pm$ 2.7a	5.7 $\pm$ 1.5b	1.9 $\pm$ 1.1b	4.2 $\pm$ 1.1b	$\chi^2_3 = 22.1$	$P < 0.001$
α -Cubebene	4.6 $\pm$ 1.7a	3.0 $\pm$ 1.2ab	0.7 $\pm$ 0.5b	0.7 $\pm$ 0.4ab	$\chi^2_3 = 9.1$	$P = 0.028$
α -Thujone	0	0.3 $\pm$ 0.3	0.1 $\pm$ 0.1	0.3 $\pm$ 0.3		
2-Carene	0	0	0	0		
Total terpenes	28 639.4 $\pm$ 1551.4a	9162.3 $\pm$ 689.0b	6432.4 $\pm$ 530.8c	8143.5 $\pm$ 667.8bc	$F_{3,236} = 109$	$P < 0.001$

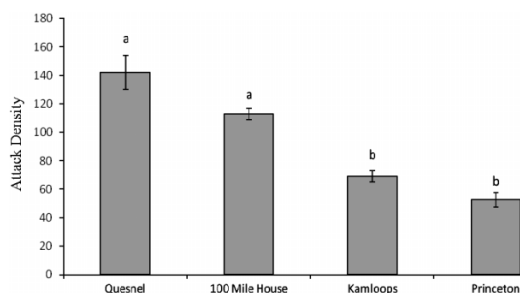
Note: Within a row a different letter indicates a significant difference between locations (Tukey's contrast or Wilcoxon's rank-sum pairwise test, $\alpha = 0.05$).

Table 3. Constitutive levels of terpenes (ppm, mean $\pm$ 1 SE) in lodgepole pine (*Pinus contorta*) trees by climatic-suitability class (CSC) and location in British Columbia (QA, Quesnel site, higher CSC; QB, Quesnel site, lower CSC; PA, Princeton site, higher CSC; PB, Princeton site, lower CSC).

Terpene	Location				χ^2_3
	QA (n = 66)	QB (n = 56)	PA (n = 59)	PB (n = 50)	
β -Phellandrene	9917.4 $\pm$ 941.1b	17869.5 $\pm$ 972.5a	7774.3 $\pm$ 601.2b	5538.9 $\pm$ 457.4c	χ^2_3 = 81.2, P < 0.001
β -Pinene	1363.7 $\pm$ 161.7b	2598.7 $\pm$ 307.1a	977.9 $\pm$ 131.1bc	623.1 $\pm$ 78.0c	$F_{3,227}$ = 17.59, P < 0.001
Δ -3-Carene	1121.9 $\pm$ 240.4b	1946.6 $\pm$ 285.7a	612.0 $\pm$ 123.4bc	517.6 $\pm$ 146.4c	χ^2_3 = 50.5, P < 0.001
Limonene	620.6 $\pm$ 128.5bc	1891.9 $\pm$ 305.9a	675.2 $\pm$ 122.8b	335.2 $\pm$ 53.1c	χ^2_3 = 53.0, P < 0.001
α -Pinene	1000.4 $\pm$ 110.0b	1366.7 $\pm$ 147.3a	610.8 $\pm$ 60.7c	363.2 $\pm$ 28.5d	χ^2_3 = 68.2, P < 0.001
Myrcene	379.3 $\pm$ 33.9b	827.0 $\pm$ 62.2a	338.3 $\pm$ 28.6b	211.4 $\pm$ 16.9c	χ^2_3 = 89.8, P < 0.001
Terpinolene	318.7 $\pm$ 49.7b	406.6 $\pm$ 40.1a	251.3 $\pm$ 32.3b	138.3 $\pm$ 19.7c	χ^2_3 = 33.5, P < 0.001
α -Phellandrene	164.9 $\pm$ 16.9b	306.4 $\pm$ 18.0a	136.2 $\pm$ 10.9b	92.6 $\pm$ 8.7c	χ^2_3 = 75.8, P < 0.001
Sabinene	211.7 $\pm$ 43.5b	260.4 $\pm$ 19.2a	121.1 $\pm$ 13.1b	78.9 $\pm$ 8.0c	χ^2_3 = 71.3, P < 0.001
Bornyl acetate	217.8 $\pm$ 49.9a	240.0 $\pm$ 81.5b	149.5 $\pm$ 62.7b	59.5 $\pm$ 5.2b	χ^2_3 = 21.4, P < 0.001
Camphene	128.4 $\pm$ 19.6b	179.9 $\pm$ 29.0a	91.7 $\pm$ 24.4b	42.3 $\pm$ 3.6c	χ^2_3 = 70.2, P < 0.001
Linalool	54.5 $\pm$ 6.1b	130.8 $\pm$ 13.6a	32.4 $\pm$ 4.1c	16.4 $\pm$ 3.3d	χ^2_3 = 91.3, P < 0.001
Terpineol	73.3 $\pm$ 9.4b	124.7 $\pm$ 10.2a	26.6 $\pm$ 4.6c	23.4 $\pm$ 5.2c	χ^2_3 = 89.0, P < 0.001
Ocimene	33.5 $\pm$ 7.7b	123.8 $\pm$ 33.1a	38.1 $\pm$ 6.7b	33.8 $\pm$ 8.6b	χ^2_3 = 21.2, P < 0.001
α -Humulene	36.7 $\pm$ 6.5b	93.9 $\pm$ 12.1a	29.3 $\pm$ 46.1b	16.7 $\pm$ 4.3b	χ^2_3 = 46.9, P < 0.001
α -Copaene	18.5 $\pm$ 4.2b	51.7 $\pm$ 6.8a	9.6 $\pm$ 2.8b	8.0 $\pm$ 2.3b	χ^2_3 = 49.7, P < 0.001
Borneol	35.1 $\pm$ 7.4a	51.6 $\pm$ 18.9a	20.8 $\pm$ 11.7b	5.6 $\pm$ 1.1b	χ^2_3 = 30.9, P < 0.001
α -Caryophyllene	19.6 $\pm$ 3.7b	46.0 $\pm$ 6.1a	14.3 $\pm$ 2.6b	8.8 $\pm$ 2.7b	χ^2_3 = 53.2, P < 0.001
Pulegone	35.1 $\pm$ 8.0ab	30.3 $\pm$ 5.4a	21.6 $\pm$ 5.1ab	8.0 $\pm$ 1.8b	χ^2_3 = 13.3, P = 0.004
<i>p</i> -Cymene	30.4 $\pm$ 4.3a	31.5 $\pm$ 4.7a	20.9 $\pm$ 4.7a	10.6 $\pm$ 2.1b	χ^2_3 = 14.27, P = 0.003
γ -Terpinene	18.4 $\pm$ 4.4b	27.1 $\pm$ 3.7a	7.7 $\pm$ 2.0bc	3.9 $\pm$ 1.6c	χ^2_3 = 33.4, P < 0.001
Camphor	27.3 $\pm$ 2.4a	15.0 $\pm$ 3.6b	24.5 $\pm$ 12.2b	2.1 $\pm$ 0.7c	χ^2_3 = 66.7, P < 0.001
α -Terpinene	12.8 $\pm$ 3.1ab	14.5 $\pm$ 2.7a	6.2 $\pm$ 1.6ab	4.2 $\pm$ 1.4b	χ^2_3 = 9.2, P = 0.027
α -Cubebene	3.9 $\pm$ 1.7	4.6 $\pm$ 1.7	2.5 $\pm$ 1.0	0.7 $\pm$ 0.4	χ^2_3 = 4.8, P = 0.184
α -Thujone	1.9 $\pm$ 0.9	0.000	0.3 $\pm$ 0.2	0.3 $\pm$ 0.3	
2-Carene	0.5 $\pm$ 0.5	0.000	0.000	0.000	
Total terpenes	15 846.3 $\pm$ 1361.1b	28 639.4 $\pm$ 1551.4a	11 993.3 $\pm$ 819.9b	8143.5 $\pm$ 667.8c	χ^2_3 = 90.8, P < 0.001

Note: A different letter indicates a significant difference between CSC and location (Tukey's contrast or Wilcoxon's rank-sum test, α = 0.05).

Fig. 1. Mountain pine beetle (*Dendroctonus ponderosae*) attack density (number of entrance holes/m²) by sampling location (Quesnel ($n = 56$), 100-Mile House ($n = 79$), Kamloops ($n = 55$), and Princeton ($n = 50$)) for lodgepole pine (*Pinus contorta*) trees in British Columbia. A different letter indicates a significant difference (ANOVA, Tukey's post-hoc test, $\alpha = 0.05$, $F_{3,236} = 19.51$, $P < 0.001$) between locations.



density; however, the levels of terpenes did not have a significant effect on pupal-chamber density (location: $F_{3,169} = 10.56$, $P < 0.001$; attack density / attack density²: $F_{1,169} = 1.86/11.17$, $P = 0.17/0.001$; DBH: $F_{1,169} = 29.92$, $P < 0.001$; data not shown). Increases in DBH and attack density were related to an increase in pupal-chamber density. There were significant differences ($F_{3,226} = 21.64$, $P < 0.001$) in attack density between sampling locations (Quesnel and Princeton) but not between CSCs at the same location. Attack densities were higher at the northernmost location (Quesnel) than at the southernmost location (Princeton) (Fig. 2). Furthermore, there was a significant effect of tree size and location and CSC on attack density when the two CSC sites at Quesnel and Princeton were compared (CSC: $F_{3,216} = 22.57$, $P < 0.001$; DBH: $F_{1,216} = 17.06$, $P < 0.001$).

Effects of terpenes on tree survival

The model best able to predict the probability of a tree surviving included level of total terpenes, tree location, attack density, and DBH (Table 5). The model is valid for trees with DBH in the 15–34 cm range and total terpene concentrations ranging from 600 to 69 900 ppm. Increasing attack density and DBH decrease the probability of tree survival, whereas increasing levels of total terpenes

increase the probability. No trees larger than DBH 25.4 cm at any of the locations survived in this experiment (Table 6). For every unit of increase in DBH and attack density, the odds of tree survival decrease by a factor of 0.78 and 0.93, respectively. However, for every unit of increase in the level of total terpenes the odds of surviving increase by a factor of 1.0001.

Discussion

Variation in terpene levels and environmental conditions

Many studies have assessed relationships between levels of resin terpenes and susceptibility of trees to attacking insects (*e.g.*, Christiansen *et al.* 1987; Klepzig *et al.* 1996; Wallin and Raffa 1999; Seybold *et al.* 2006). Lodgepole pine occupies a large geographical range in North America in which it interacts with many potentially damaging agents, including mountain pine beetle. Thus, geographical differences in constitutive resin chemistry relevant to mountain pine beetle attack may have arisen because of a diverse array of complex co-evolutionary interactions (Thompson 1997). We found significant geographical differences in the levels of various terpenes in lodgepole pine. This was particularly true of the northernmost site (Quesnel), where we found significantly higher concentrations of most of the terpenes tested, including Δ -3-carene, limonene, myrcene, α - and β -pinene, β -phellandrene, terpinolene, and total terpenes, than at the other sites. In British Columbia, most mountain pine beetle outbreaks occur in the southern half of the province (Carroll *et al.* 2004). Limonene, Δ -3-carene, and α -pinene are lethal to mountain pine beetle eggs (Raffa and Berryman 1983a) and toxic to adult western pine beetles (Smith 1965). These geographical differences in terpene profiles may be due in part to the level of historical interaction between these lodgepole pine populations and the mountain pine beetle.

Sturgeon (1979) found that ponderosa pine had higher levels of limonene in the presence of western pine beetle (limonene is toxic to the western pine beetle) as well as higher concentrations of myrcene (used by the beetle to synergize its pheromone). The variation in

Table 4. Effect of location, tree size, and limonene level on attack density (number of entrance holes/m²) for mountain pine beetle (*Dendroctonus ponderosae*) on lodgepole pine (*Pinus contorta*) trees in British Columbia.

Variable	Value*	F	df	P
Location†		19.29	3,225	
Quesnel	52.44 (10.67)			0.0041
100 Mile House	83.42 (27.90)			0.0031
Kamloops	38.13 (8.46)			<0.001
Princeton	31.82 (8.55)			<0.001
Tree size (cm)	3.61 (0.86)	18.15	1,225	<0.001
Limonene (ppm)	0.0057 (0.0022)	6.75	1,225	0.01

*Mean with SE in parentheses.

†The location of interest is used as the intercept in an ANCOVA model.

terpene profiles among our test locations also indicates that there is potential for variation in the interaction between the mountain pine beetle and its host tree, which we did observe in the differences in attack density, though the connection between attack density and all of the terpenes was not significant.

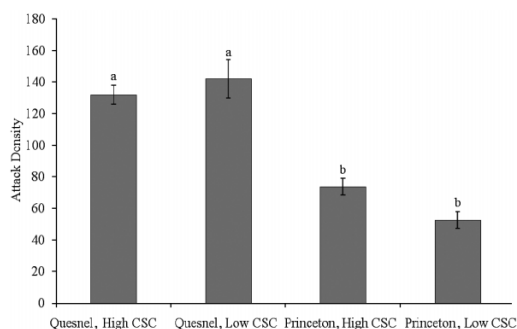
The lower CSC site near Quesnel generally had significantly higher concentrations of most of the terpenes, including all seven predominant terpenes, than did the other sites. However, the lower CSC site near Princeton generally had significantly lower concentrations of most terpenes, including all seven predominant terpenes, than did the other sites. Differences between these sites are doubtless due to environmental differences, in addition to historical beetle pressure. Because defenses require an investment of resources (Herms and Mattson 1992), differences in levels of defense against bark beetles and their associated fungi, and environmental stressors, will be linked to factors that affect photosynthetic efficiency (Christiansen *et al.* 1987) and to other nutrient limitations (Gilmore 1977; Waring and Pitman 1985; Lewinsohn *et al.* 1993). For instance, red pine (*Pinus resinosa* Aiton) seedlings stressed by shading were found to have lower levels of induced chemical changes when challenged by a bark beetle vectored fungus (Klepzig *et al.* 1995). Thus, terpene composition varies across a geographical range, not only because of historical beetle pressure or other interactions with damaging agents, but also because of environmental differences between locations. Environment conditions and historical beetle pressure are also linked

because ectothermic insects are highly dependent on temperature and other climate variables (Carroll *et al.* 2004). In fact, the variables used to determine the historical likelihood of beetle success, which in turn determined the CSCs used in these analyses, are based on historical climate data (Carroll *et al.* 2004).

Effects of terpenes on attack density and beetle reproductive success

Although resin terpene levels in phloem varied by location (Table 2), only limonene levels were found to significantly affect attack density (Table 4). Increases in limonene level were related to an increase in attack density, which was an unexpected result. Limonene was toxic to the southern pine beetle (*Dendroctonus frontalis* Zimmermann) in a laboratory bioassay (Coyne and Lott 1976). In addition, among compounds tested by Smith (1965) against western pine beetle, limonene vapor was the most toxic monoterpene. Therefore, we expected that higher limonene levels would result in lower levels of beetle attack. Limonene level was not a significant variable in the linear mixed-effects model used to analyze differences in attack density between CSCs at the two different locations, despite the significant difference in levels present in resin. It is possible that limonene, despite its toxicity, was an insufficient deterrent to the beetles at these extreme attack densities. In other words, the correlation between increased limonene concentration and increased attack density is not a causal relationship. Alternatively, the increase in attack density could be an indication that a higher

Fig. 2. Mountain pine beetle (*Dendroctonus ponderosae*) attack density (entrance holes/m²) by sampling location (Quesnel ($n = 66, 56$), Princeton ($n = 58, 50$)) and climatic-suitability class (CSC; high and low) for lodgepole pine (*Pinus contorta*) trees in British Columbia. A different letter indicates a significant difference (ANOVA, Tukey's post-hoc test, $\alpha = 0.05$, $F_{3,226} = 21.64$, $P < 0.001$) between locations.



attack density was required to successfully overcome a stronger host defense.

It is surprising that the α -pinene level was not a significant variable in the model in relation to attack density, because α -pinene serves as a precursor to, and has synergistic properties with, the aggregation pheromone *trans*-verbenol (Pitman 1971; Hughes 1973; Conn *et al.* 1984). It can also, however, be auto-oxidized to verbenone, an anti-aggregation pheromone (Borden *et al.* 1987; Hunt *et al.* 1989). Higher levels of α -pinene allow the mountain pine beetle to optimize attack dynamics on host trees (Raffa and Berryman 1983b) by providing more pheromone precursors to attacking beetles. In areas of high beetle population pressure, such an effect would reduce the attack density required to kill an individual tree, reducing intraspecific competition on such trees. Levels of α -pinene were not found to be a significant factor for tree survival, indicating that on its own, α -pinene did not reduce attack to the point of allowing tree survival. The complex relationship between mountain pine beetle and α -pinene may explain why a clear relationship between α -pinene and attack density could not be found in a comparison of all locations.

Individual terpenes did not influence beetle reproductive success as measured by the density of pupal chambers. Raffa and Berryman

(1983b) found that at high attack density (approximately 80 galleries/m²) the host does not have a suppressive effect on brood development (mean attack densities (number of entrance holes/m²) observed in our study were higher than this). However, DBH, location, and original attack density did relate to pupal-chamber density. Total attacks and brood production have been positively correlated with tree diameter (Safranyik and Carroll 2006), which is consistent with our findings. Attack density was strongly correlated with the number of pupal chambers, but not in a linear way. This is because once the optimum attack density has been exceeded, intraspecific competition is the main factor that would reduce larval success and, ultimately, total pupal chambers (Raffa and Berryman 1983b).

Variation in resin terpene levels and tree survival

A large DBH was correlated with an increase in the probability of tree survival the following spring in all areas that we sampled. Because the sampling for tree survival yielded a reduced data set, owing to loss of plots (Table 6), it is possible that the analysis misrepresents some dynamics that may occur, particularly in the northern area, where the sample size was most reduced. However, this analysis does serve to indicate what may be happening in stands across a large portion of the range of lodgepole pine. Many studies have demonstrated that large-diameter lodgepole pines are generally the first trees in a stand to be attacked and killed (*e.g.*, Cole and Amman 1969; Safranyik *et al.* 1974). Phloem thickness, which is related to tree size, has been correlated with beetle success (Amman 1972; Amman and Pace 1976; Berryman 1976). Cole *et al.* (1981) found that the resin monoterpene concentration is higher in trees with thicker phloem, which also relates terpene level to tree size (Amman and Cole 1983), although a significant linear relationship in this data set with four locations was not found. Higher levels of total terpenes did relate to an increase in the probability of tree survival, indicating that high levels may protect the tree once it has been attacked. Although terpenes are often viewed as defense compounds, bark beetles also use such compounds as kairomones to

Table 5. Effect of location, tree size, mountain pine beetle (*Dendroctonus ponderosae*) attack density, and level of total terpenes on the probability of survival of lodgepole pine (*Pinus contorta*) trees in British Columbia.

Variable	Estimate	SE	P
Location*			
Quesnel	1.32	1.65	0.43
100 Mile House	2.23	2.14	0.30
Kamloops	2.48	1.17	0.034
Princeton	2.20	1.18	0.061
Tree size (cm)	-0.24	0.10	0.020
Attack density (entrance holes/m ²)	-6.93×10^{-2}	1.49×10^{-2}	> 0.001
Total terpenes (ppm)	1.01×10^{-4}	4.66×10^{-5}	0.031

*The location of interest is used as the intercept in the model.

identify and locate suitable hosts (Conn *et al.* 1983). Thus, lodgepole pines with high levels of resin terpenes (as well as likely being bigger and more perceptible) are more apparent to foraging mountain pine beetles.

There is selective pressure on mountain pine beetles to efficiently locate suitable hosts. They must find susceptible hosts in stands that also contain nonhosts and resistant hosts (Atkins 1966). In addition, they have a limited energy supply (Bell 1990), and by minimizing foraging time, they minimize the time that they are exposed to other hazards such as inclement weather or predators and parasitoids (Dahlsten 1982). Pioneering beetles may locate hosts using visual cues (Shepherd 1966; Cole and Amman 1969) or at random (Hynum and Berryman 1980) and then test the potential host for suitability after landing on it (Elkinton and Wood 1980; Wallin and Raffa 2000; Pureswaran and Borden 2003). But they also likely utilize chemical cues for host selection while in flight. Moeck and Simmons (1991) found that mountain pine beetles are attracted to host odors, even in the absence of aggregation pheromone or visual cues. Like other bark beetles, this species has the ability to detect and avoid host and nonhost volatiles in flight (Huber *et al.* 2003). Pureswaran *et al.* (2004) found that there were enough quantitative differences in the volatile monoterpene profiles of several conifer species that such volatiles could be used for host identification by flying insects. As in their study, we found higher levels of α -pinene in locations farther north (though we did not find significant differences between

Princeton and 100 Mile House). We also found more quantitative differences in terpenes than did Pureswaran *et al.* (2004), which may indicate that there is more variation in populations of trees in each location than we were able to detect in our study. Trees in the south of British Columbia, with greater previous exposure to mountain pine beetle outbreaks, seem to have been selected to reduce their apparency to the beetle by maintaining low concentrations of terpenes compared with those sampled in the northern area. Supporting this is the fact that the lower CSC in Quesnel (the northernmost site) had significantly higher levels of all seven predominant terpenes than the higher CSC in the same location. If there had been historical mountain pine beetle outbreaks near Quesnel they would most likely have occurred in the areas that were climatically more suitable for the beetle. However, the lower CSC in Princeton had the lowest levels of the seven predominant terpenes, which is not consistent with this theory. This is explained by Princeton's location in the center of the historical beetle range in British Columbia. During outbreaks, beetles are often pushed into climatically less suitable areas, and high tree mortality can occur even in areas classified in the lower CSCs (Carroll *et al.* 2004). This means that any differences in terpene levels due to historical beetle pressure would be more pronounced near the edges of the historical range of the beetle, such as the northern location, when climatically suitable regions, such as the lower CSC site near Quesnel and the lower CSC site near Princeton, are compared.

Table 6. Numbers of British Columbia lodgepole pine (*Pinus contorta*) trees in four size ranges that were alive or dead, based upon the foliage in the spring following attack by mountain pine beetle (*Dendroctonus ponderosae*).

Location	DBH 15–20.4 cm		DBH 20.5–25.4 cm		DBH 25.5–29.4 cm		DBH > 29.5 cm	
	Alive	Dead	Alive	Dead	Alive	Dead	Alive	Dead
Quesnel	9	6	2	5	0	0	0	0
100 Mile House	1	47	0	16	0	9	0	6
Kamloops	11	26	1	12	0	3	0	2
Princeton	14	19	2	11	0	2	0	0

Note: No surviving trees at Quesnel or 100 Mile House were attacked. All 12 surviving trees at Kamloops and 8 surviving trees at Princeton were attacked.

In an analogous situation, Tomlin *et al.* (1997) found that four clones of Sitka spruce, *Picea sitchensis* (Bong.) Carr. (Pinaceae), classified as resistant to the white pine weevil, *Pissodes strobi* Peck (Coleoptera: Curculionidae), had significantly lower levels of 10 terpenes than did clones classified as susceptible. Their results indicated that there were multiple resistant chemotypes, based on the terpene profiles, but that the resistance could be classified as due either to repellency or to a lack of apparency. In addition, the sympatric presence of other suitable mountain pine beetle hosts (*e.g.*, ponderosa pine; Wood 1982) could also create selective benefits for trees that are not perceived by the beetles. In the stands that we sampled, individual lodgepole pines with reduced levels of constitutive terpenes (and thus more difficult for foraging beetles to locate) may be better off because other, potentially more apparent host species are available. This apparency strategy is likely more effective in preventing attack than is the production of sufficient constitutive defenses, and may be a better use of energy resources for lodgepole pines growing among other suitable host species. These variations in ecosystems, resource availability, alternative hosts, and a variety of other competitors for resources across this geographical range could lead to different selective pressures in different populations of mountain pine beetle and lodgepole pine (Thompson 1997), which would result in this variation in terpene profiles between locations.

All of our experiments were performed in the midst of a population outbreak, therefore

studies designed to examine similar defense variables at different stages in the population cycle would expand on the relationship between the mountain pine beetle and its lodgepole pine host. During times of lower beetle population pressure, host-tree defenses, such as terpenes, could play a larger role in beetle attack dynamics. Wallin and Raffa (2004) showed that epidemic populations of the spruce beetle (*Dendroctonus rufipennis* Kirby), another eruptive bark beetle, exhibited a decreased avoidance of high concentrations of α -pinene with an increase in numbers of conspecifics. In addition, an examination of the particular concentrations of terpenes necessary to affect successful host location, maintenance of optimum attack density, and reproduction of mountain pine beetles could clarify the importance of the levels of particular terpenes in the phloem resin. For example, it would be interesting to determine the levels of α -pinene (utilized by the mountain pine beetle for producing aggregation and anti-aggregation pheromones and also potentially toxic to colonizing beetles) that allow the beetles to maintain optimum attack density but avoid the toxic effects of the compound.

Acknowledgements

We thank the Natural Resources Canada Canadian Forest Service Mountain Pine Beetle Initiative (8.45), the Natural Sciences and Engineering Research Council of Canada, the Canada Research Chairs Program, and the Canada Foundation for Innovation for financial

support of this project. We thank Dr. B. Staffan Lindgren for advice and input. We also thank Mr. Marco Hernandez, Ms. Anna Milner, Mr. Daniel Ott, Ms. Caitlin Pitt, and Mr. Jeff Selesnic for assistance with fieldwork. Analytical chemistry on phloem samples was performed by Mr. Clive Dawson of the British Columbia Ministry of Forests and Range Research Branch.

References

- Amman, G.D. 1972. Mountain pine beetle brood production in relation to thickness of lodgepole pine phloem. *Journal of Economic Entomology*, **65**: 138–140.
- Amman, G.D., and Cole, W.E. 1983. Mountain pine beetle dynamics in lodgepole pine forests. Part II: Population dynamics. United States Forest Service General Technical Report INT-145.
- Amman, G.D., and Pace, V.E. 1976. Optimum egg gallery densities for the mountain pine beetle in relation to lodgepole pine phloem thickness. United States Forest Service Research Note INT-209.
- Atkins, M.D. 1966. Behavioral variation among scolytids in relation to their habitat. *The Canadian Entomologist*, **98**: 285–288. doi:10.4039/Ent98285-3.
- Bell, W.J. 1990. Searching behaviour: the behavioural ecology of finding resources. Chapman and Hall, New York.
- Berryman, A.A. 1976. Theoretical explanation of mountain pine beetle dynamics in lodgepole pine forests. *Environmental Entomology*, **5**: 1225–1233.
- Borden, J.H. 1985. Aggregation pheromones. *In* Comprehensive insect physiology, biochemistry, and pharmacology. Edited by G.A. Kerkut and L.I. Gilbert. Pergamon Press, Oxford, United Kingdom. pp. 257–285.
- Borden, J.H., Ryker, L.J., Chong, L.J., Pierce, F., H.D., Johnston, B.D., and Oehlschlager, A.C. 1987. Response of the mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae), to five semiochemicals in British Columbia lodgepole pine forests. *Canadian Journal of Forest Research*, **17**: 118–128. doi:10.1139/x87-023.
- Carroll, A.L., Taylor, S.W., Régnière, J., and Safranyik, L. 2004. Effects of climate change on range expansion by the mountain pine beetle in British Columbia. *In* Proceedings of the Mountain Pine Beetle Symposium: Challenges and Solutions, Kelowna, British Columbia, 30–31 October 2003. Edited by T.L. Shore, J.E. Brooks, and J.E. Stone. Natural Resources Canada, Victoria, British Columbia. pp. 223–232.
- Christiansen, E., Waring, R.H., and Berryman, A.A. 1987. Resistance of conifers to bark beetle attack: searching for general relationships. *Forest Ecology and Management*, **22**: 89–106. doi:10.1016/0378-1127(87)90098-3.
- Cole, W.E., and Amman, G.D. 1969. Mountain pine beetle infestations in relation to lodgepole pine diameters. United States Forest Service. INT-95.
- Cole, W.E., Guymon, E.P., Jensen, C.E. 1981. Monoterpenes of lodgepole pine phloem as related to mountain pine beetles. United States Forest Service Research Paper INT-281.
- Conn, J.E., Borden, J.H., Scott, B.E., Friskie, L.M., Pierce, Jr., H.D., and Oehlschlager, A.C. 1983. Semiochemicals for the mountain pine beetle, *Dendroctonus ponderosae* (Coleoptera: Scolytidae) in British Columbia: field trapping studies. *Canadian Journal of Forest Research*, **13**: 320–324. doi:10.1139/x83-045.
- Conn, J.E., Borden, J.H., Hunt, D.W.A., Holman, J., Whitney, H.S., Spanier, O.J., Pierce, Jr., H.D., and Oehlschlager, A.C. 1984. Pheromone production by axenically reared *Dendroctonus ponderosae* and *Ips paraconfusus* (Coleoptera: Scolytidae). *Journal of Chemical Ecology*, **10**: 281–290. doi:10.1007/BF00987856.
- Coyne, J.F., and Lott, L.H. 1976. Toxicity of substances in pine oleoresin to southern pine beetles. *The Journal of the Georgia Entomological Society*, **11**: 297–301.
- Craighead, F.C., Miller, J.M., Evenden, J.C., and Keen, F.P. 1931. Control work against bark beetles in western forests and an appraisal of its results. *Journal of Forestry*, **29**: 1001–1018.
- Dahlsten, D.L. 1982. Relationship between bark beetles and their natural enemies. *In* Bark beetles in North American conifers: a system for the study of evolutionary biology. Edited by J.B. Mitton and K.B. Sturgeon. University of Texas Press, Austin, Texas. pp. 140–182.
- Elkinton, J.S., and Wood, D.L. 1980. Feeding and boring behavior of the bark beetle *Ips paraconfusus* (Coleoptera: Scolytidae) on the bark of a host and its non-host tree species. *The Canadian Entomologist*, **112**: 797–809. doi:10.4039/Ent112797-8.
- Forrest, G.I. 1980. Geographical variation in the monoterpenes of *Pinus contorta* oleoresin. *Biochemical Systematics and Ecology*, **8**: 343–359. doi:10.1016/0305-1978(80)90037-X.
- Franceschi, V.R., Krokene, P., Christiansen, E., and Krekling, T. 2005. Anatomical and chemical defenses of conifer bark against bark beetles and other pests. *New Phytologist*, **167**: 353–376. PMID: 15998390 doi:10.1111/j.1469-8137.2005.01436.x.
- Franklin, E.C., and Snyder, E.B. 1971. Variation and inheritance of monoterpene composition in longleaf pine. *Forest Science*, **17**: 178–179.
- Gershenzon, J., and Croteau, R.B. 1991. Terpenoids. *In* Herbivores: their interactions with secondary plant metabolites. Edited by G.A. Rosenthal and M.R. Berenbaum. Academic Press, San Diego, California. pp. 165–219.

- Gilmore, A.R. 1977. Effects of soil moisture stress on monoterpenes in loblolly pine. *Journal of Chemical Ecology*, **3**: 667–676. doi:10.1007/BF00988066.
- Hanover, J.W. 1966. Genetics of terpenes: I. Gene control of monoterpene levels in *Pinus monticola* Dougl. *Heredity*, **21**: 73–84. doi:10.1038/hdy.1966.5.
- Hanover, J.W., and Furniss, M.M. 1966. Monoterpene concentration in Douglas-fir in relation to geographic location and resistance to attack by the Douglas-fir beetle. In *Joint Proceedings of the Second Genetics Workshop of the Society of American Foresters and the Seventh Lake States Forest Tree Improvement Conference*. United States Forest Service Research Paper NC-6. pp. 23–28.
- Hermes, D.A., and Mattson, W.J. 1992. The dilemma of plants: to grow or defend. *The Quarterly Review of Biology*, **67**: 283–335. doi:10.1086/417659.
- Huber, D.P.W., Gries, R., Borden, J.H., and Pierce, H.D., Jr. 2003. A survey of antennal responses by five species of coniferophagous bark beetles (Coleoptera: Scolytidae) to bark volatiles of six species of angiosperm trees. *Chemoecology*, **10**: 103–113. doi:10.1007/PL00001811.
- Huber, D.P.W., Ralph, S., and Bohlmann, J. 2004. Genomic hardwiring and phenotypic plasticity of terpenoid-based defenses in conifers. *Journal of Chemical Ecology*, **30**: 2399–2418. PMID:15724963 doi:10.1007/s10886-004-7942-2.
- Hughes, P.R. 1973. Effect of α -pinene exposure on *trans*-verbenol synthesis in *Dendroctonus ponderosae* Hopk. *Naturwissenschaften*, **60**: 261–262. doi:10.1007/BF00625726.
- Hunt, D.W.A., Borden, J.H., Lindgren, B.S., and Gries, G. 1989. The role of autoxidation of α -pinene in the production of pheromones of *Dendroctonus ponderosae* (Coleoptera: Scolytidae). *Canadian Journal of Forest Research*, **19**: 1275–1282.
- Hynum, B.G., and Berryman, A.A. 1980. *Dendroctonus ponderosae* (Coleoptera: Scolytidae): Pre-aggregation landing and gallery initiation on lodgepole pine. *The Canadian Entomologist*, **112**: 185–191. doi:10.4039/Ent112185-2.
- Keeling, C.I., and Bohlmann, J. 2006. Genes, enzymes and chemicals of terpenoid diversity in the constitutive and induced defense of conifers against insects and pathogens. *New Phytologist*, **170**: 657–675. PMID:16684230 doi:10.1111/j.1469-8137.2006.01716.x.
- Klepzig, K.D., Kruger, E.L., Smalley, E.B., and Raffa, K.F. 1995. Effects of biotic and abiotic stress on induced accumulation of terpenes and phenolics in red pines inoculated with bark beetle-vectored fungus. *Journal of Chemical Ecology*, **21**: 601–626. doi:10.1007/BF02033704.
- Klepzig, K.D., Smalley, E.B., and Raffa, K.F. 1996. Combined chemical defenses against an insect-fungal complex. *Journal of Chemical Ecology*, **22**: 1367–1388. doi:10.1007/BF02027719.
- Lee, S., Kim, J.-J., and Breuil, C. 2006. Pathogenicity of *Leptographium longiclavatum* associated with *Dendroctonus ponderosae* to *Pinus contorta*. *Canadian Journal of Forest Research*, **36**: 2867–2872. doi:10.1139/X06-194.
- Lewinsohn, E., Gijzen, M., Muzika, R.M., Barton, K., and Croteau, R. 1993. Oleoresinosis in grand fir (*Abies grandis*) saplings and mature trees. *Plant Physiology*, **101**: 1021–1028. PMID:12231755.
- Logan, J.A., and Powell, J.A. 2001. Ghost forests, global warming, and the mountain pine beetle (Coleoptera: Scolytidae). *American Entomologist*, **47**: 160–172.
- Mickey, R.M., Dunn, O.J., and Clark, V.A. 2004. Applied statistics analysis of variance and regression. John Wiley & Sons, Inc., Hoboken, New Jersey.
- Mitton, J.B., and Sturgeon, K.B. 1982. Biotic interactions and evolutionary change. In *Bark beetles in North American conifers: a system for the study of evolutionary biology*. Edited by J.B. Mitton and K.B. Sturgeon. University of Texas Press, Austin, Texas. pp. 3–20.
- Moeck, H.A., and Simmons, C.S. 1991. Primary attraction of mountain pine beetle, *Dendroctonus ponderosae* Hopk. (Coleoptera: Scolytidae), to bolts of lodgepole pine. *The Canadian Entomologist*, **123**: 299–304. doi:10.4039/Ent123299-2.
- Pitman, G.B. 1971. *Trans*-verbenol and α -pinene: their utility in manipulation of the mountain pine beetle. *Journal of Economic Entomology*, **64**: 426–430.
- Pureswaran, D.S., and Borden, J.H. 2003. Test of semiochemical mediated host specificity in four species of tree killing bark beetles (Coleoptera: Scolytidae). *Environmental Entomology*, **32**: 963–969. doi:10.1603/0046-225X-32.5.963.
- Pureswaran, D.S., Gries, R., and Borden, J.H. 2004. Quantitative variation in monoterpenes in four species of conifers. *Biochemical Systematics and Ecology*, **32**: 1109–1136. doi:10.1016/j.bse.2004.04.006.
- Pureswaran, D.S., Sullivan, B.T., and Ayres, M.P. 2006. Fitness consequences of pheromone production and host selection strategies in a tree-killing bark beetle (Coleoptera: Curculionidae: Scolytinae). *Oecologia*, **148**: 720–728. doi:10.1007/s00442-006-0400-9.
- R Development Core Team. 2008. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0. Available from <http://www.R-project.org> [accessed September 2008].
- Raffa, K.F., and Berryman, A.A. 1982. Physiological differences between lodgepole pines resistant and susceptible to the mountain pine beetle and associated microorganisms. *Environmental Entomologist*, **11**: 486–492.
- Raffa, K.F., and Berryman, A.A. 1983a. Physiological aspects of lodgepole pine wound responses to a fungal symbiont of the mountain pine beetle, *Dendroctonus ponderosae* (Coleoptera: Scolytidae). *The Canadian Entomologist*, **115**: 723–734. doi:10.4039/Ent115723-7.

- Raffa, K.F., and Berryman, A.A. 1983b. The role of host plant resistance in the colonization behavior and ecology of bark beetles (Coleoptera: Scolytidae). *Ecological Monographs*, **53**: 27–49. doi:10.2307/1942586.
- Raffa, K.F., and Berryman, A.A. 1987. Interacting selective pressures in conifer – bark beetle systems: a basis for reciprocal adaptations? *The American Naturalist*, **129**: 234–262. doi:10.1086/284633.
- Raffa, K.F., Aukema, B.H., Bentz, B.J., Carroll, A.L., Hicke, J.A., Turner, M.G., and Romme, W.H. 2008. Cross-scale drivers of natural disturbances prone to anthropogenic amplification: the dynamics of bark beetle eruptions. *BioScience*, **58**: 501–517. doi:10.1641/B580607.
- Reid, R.W., Whitney, H.S., and Watson, J.A. 1967. Reactions of lodgepole pine to attack by *Dendroctonus ponderosae* Hopkins and blue stain fungi. *Canadian Journal of Botany*, **45**: 1115–1126. doi:10.1139/b67-116.
- Rockwood, D.L. 1973. Variation in the monoterpene composition of two oleoresin systems of loblolly pine. *Forest Science*, **19**: 147–153.
- Rudinsky, J.A. 1962. Ecology of Scolytidae. *In* Annual review of entomology. *Edited by* E.A. Steinhaus and R.F. Smith. Annual Reviews, Inc., Palo Alto, California. pp. 327–348.
- Safranyik, L., and Carroll, A.L. 2006. The biology and epidemiology of the mountain pine beetle in lodgepole pine forests. *In* The mountain pine beetle: a synthesis of biology, management, and impacts on lodgepole pine. *Edited by* L. Safranyik and W.R. Wilson. Pacific Forestry Centre, Canadian Forest Service, Natural Resources Canada, Victoria, British Columbia. pp. 3–66.
- Safranyik, L., Shrimpton, D.M., and Whitney, H.S. 1974. Management of lodgepole pine to reduce losses from the mountain pine beetle. Victoria, BC. Technical Report No. 1, Pacific Forestry Centre, Canadian Forest Service, Natural Resources Canada, Victoria, British Columbia.
- Seybold, S.J., Huber, D.P.W., Lee, J.C., Graves, A.D., and Bohlmann, J. 2006. Pine monoterpenes and pine bark beetles: a marriage of convenience for defense and chemical communication. *Phytochemistry Reviews*, **5**: 143–178. doi:10.1007/s11101-006-9002-8.
- Shepherd, R.F. 1966. Factors influencing the orientation and rates of activity of *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae). *The Canadian Entomologist*, **98**: 507–518. doi:10.4039/Ent98507-5.
- Smith, R.H. 1963. Toxicity of pine resin vapors to three species of *Dendroctonus* bark beetles. *Journal of Economic Entomology*, **56**: 827–831.
- Smith, R.H. 1965. Effect of monoterpene vapors on the western pine beetle. *Journal of Economic Entomology*, **58**: 509–510.
- Smith, R.H. 1975. Formula for describing effect of insect and host tree factors on resistance to western pine beetle attack. *Journal of Economic Entomology*, **68**: 841–844.
- Smith, R.H. 1983. Monoterpenes of lodgepole pine xylem resin: a regional study in western United States. *Forest Science*, **29**: 333–340.
- Squillace, A.E. 1971. Inheritance of monoterpene composition in cortical oleoresin of slash pine. *Forest Science*, **17**: 381–387.
- Sturgeon, K.B. 1979. Monoterpene variation in ponderosa pine xylem resin related to western pine beetle predation. *Evolution*, **33**: 803–814. doi:10.2307/2407647.
- Tabachnick, B.G., and Fidell, L.S. 2001. Using multivariate statistics. 4th ed. Allyn and Bacon, Boston, Massachusetts.
- Taylor, S.W., and Carroll, A.L. 2004. Disturbance, forest age, and mountain pine beetle outbreak dynamics in BC: a historical perspective. *In* Mountain Pine Beetle Symposium: Challenges and Solutions, Kelowna, British Columbia, 30–31 October 2003. *Edited by* T.L. Shore, J.E. Brooks, and J.E. Stone. Pacific Forestry Centre, Canadian Forest Service, Natural Resources Canada, Victoria, British Columbia. pp. 41–51.
- Thompson, J.N. 1997. Evaluating the dynamics of coevolution among geographically structured populations. *Ecology*, **78**: 1619–1623. doi:10.1890/0012-9658(1997)078[1619:ETDOCA]2.0.CO;2.
- Tomlin, E.S., Borden, J.H., and Pierce, H.D., Jr. 1997. Relationship between volatile foliar terpenes and resistance of Sitka spruce to the white pine weevil. *Forest Science*, **43**: 501–508.
- Wallin, K.F., and Raffa, K.F. 1999. Altered constitutive and inducible phloem monoterpenes following natural defoliation of jack pine: implications to host mediated interguild interactions and plant defense theories. *Journal of Chemical Ecology*, **25**: 861–880. doi:10.1023/A:1020853019309.
- Wallin, K.F., and Raffa, K.F. 2000. Influences of host chemicals and internal physiology on the multiple steps of postlanding host acceptance behavior of *Ips pini* (Coleoptera: Scolytidae). *Environmental Entomology*, **29**: 442–453. doi:10.1603/0046-225X-29.3.442.
- Wallin, K.F., and Raffa, K.F. 2004. Feedback between individual host selection behavior and population dynamics in an eruptive herbivore. *Ecological Monographs*, **74**: 101–116. doi:10.1890/02-4004.
- Waring, R.H., and Pitman, G.B. 1985. Modifying lodgepole pine stands to change susceptibility to mountain pine beetle attack. *Ecology*, **66**: 889–897. doi:10.2307/1940551.
- Westfall, J., and Ebata, T. 2008. Summary of forest health conditions in British Columbia. Pest Management Report No. 15, British Columbia Ministry of Forests and Range, Victoria, British Columbia.
- Wood, S.L. 1982. The bark and ambrosia beetles of North and Central America (Coleoptera: Scolytidae): a taxonomic monograph. Great Basin Naturalist Memoirs No. 6, Brigham Young University, Provo, Utah.

Appendix A

Table A1. Correlation matrix for DBH and the seven predominant terpenes in lodgepole pine (*Pinus contorta*) trees in British Columbia. Note that a bivariate correlation creates multicollinearity when it is above 0.90 as defined by Tabachnick and Fidell (2001).

	DBH	Δ -3-Carene	Limonene	Myrcene	β -Phellandrene	α -Pinene	β -Pinene	Terpinolene
DBH	—	−0.02	−0.01	−0.03	−0.01	−0.02	0.01	−0.01
Δ -3-Carene	−0.02	—	0.06	0.49	0.47	0.30	0.21	0.73
Limonene	−0.01	0.06	—	0.49	0.48	0.35	0.42	0.35
Myrcene	−0.03	0.49	0.49	—	0.87	0.64	0.63	0.62
β -Phellandrene	−0.01	0.47	0.48	0.87	—	0.72	0.68	0.65
α -Pinene	−0.02	0.30	0.35	0.64	0.72	—	0.63	0.40
β -Pinene	0.01	0.21	0.42	0.63	0.68	0.63	—	0.34
Terpinolene	−0.01	0.73	0.35	0.62	0.65	0.40	0.34	—